Culturally-Constrained AI Generation of Hmong Ethnic Patterns for Digital Heritage Preservation

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Abstract. This paper introduces an AI-driven framework for generating traditional Hmong textile patterns with embedded cultural constraints, ensuring both aesthetic fidelity and symbolic authenticity. Unlike conventional generative models that overlook nuanced cultural rules, our three-stage architecture integrates: (1) Cultural Feature Encoding, which fuses visual structure with semantic metadata; (2) Controlled Generative Modeling using Stable Diffusion fine-tuned via LoRA; and (3) Multi-tier Cultural Validation to verify pattern, symbol, and geometric integrity. Trained on a curated dataset of 55 original Hmong motifs—augmented to 550 culturally-consistent samples—our proposed method improves generative quality over the baseline, achieving markedly lower Fréchet Inception Distance (FID), strong Cultural Consistency Score (CCS = 86%), and high validation performance across patterns, symbols, and structural criteria. We contribute the first integrated pipeline for culturally-grounded pattern generation, a publicly available annotated Hmong dataset, an automated validation suite, and a complete open-source implementation. The code is available at https://github.com/ThanhTrunggDEV/GenAI

Keywords: Generative AI, cultural heritage preservation, Stable Diffusion, Hmong textiles, constrained generation, LoRA fine-tuning, computational ethnography.

1 Introduction

Traditional costumes of every ethnic group in the world, as well as in Vietnam, have their own unique characteristics, reflecting the cultural and historical beauty of that ethnic group. Vietnam has 54 ethnic groups, among which the Hmong are a unique and distinctive group, and these characteristics are expressed in each costume with different patterns and motifs. Even from a distance, we can identify a Hmong costume through its patterns and motifs. The textile motifs of the Hmong community in Vietnam constitute a living repository of intangible cultural heritage, encoding cosmological beliefs, social identity, and ancestral memory [1]. Yet this tradition faces erosion due to generational discontinuity, economic marginalization, and the dwindling number of master artisans. Recent advances in generative artificial intelligence offer transformative potential for digital preservation, but models such as Stable Diffusion [2] often lack mechanisms to enforce culturally-specific constraints, risking misrepresentation or symbolic dilution.

This work bridges the gap between generative AI and cultural fidelity by proposing a constrained generation framework for Hmong patterns. We identify three core challenges: Scarcity of culturally annotated data for ethnic motifs. Complexity of encoding cultural rules (color symbolism, ritual context, geometric proportionality). Absence of automated cultural validation metrics beyond visual similarity. Our contributions address these gaps through a novel three-stage pipeline—the first integrated framework to jointly couple cultural feature encoding, LoRA-based constrained diffusion, and automated multi-tier validation for ethnic minority textile pattern synthesis. The first structured, culturally annotated dataset of Hmong textile patterns. An automated validator assessing pattern, symbol, and structural integrity. Full open-source release of code, models, and data to support reproducibility and community engagement.

2 Related Work

Traditional costumes of each ethnic group play a very important role, reflecting the unique characteristics of each group, contributing to the preservation and promotion of their value. In recent years, many scientists and managers around the world and in Vietnam have shown interest in researching and proposing solutions to apply virtual reality (VR), augmented reality (AR), and artificial intelligence (AI) technologies to support the digitization, presentation, and creation of patterns, motifs, and costumes. Specifically: In 2025, the research group of Ma et al. [3] proposed a three-tier framework for the experience of exploring ethnic costumes, combining AI content generation and personalized interaction, to help users not only observe but also create costume images based on the cultural genes of each ethnic group. Sangamuang et al. [4] developed a virtual reality/Metaverse environment combining AI generation and virtual assistants to preserve and promote the Lamphun Brocade textile heritage (Thailand). The model shows that AI can support real-time cultural interpretation and create a deeply interactive experience. In addition, there are studies that integrate AI virtual assistants into VR museum environments to increase interaction and learning about textile culture. AI not only provides information but also supports multilingual and personalized experiences, allowing users to explore the technical details and historical context of traditional textile patterns in greater depth [5]. There is research on automatically extracting and generating traditional clothing pattern information from images using the RSKP-UNet model combined with attention and stable dif-fusion model [6], in which AI application in feature segmentation and new pattern generation paves the way for systematic digitization of traditional clothing pattern samples. Authors Shirin Hajahmadi et al. [7] analyzed how VR and AI are used as storytelling tools in the digital environment of traditional fashion, highlighting the potential of virtual reality in recreating and presenting fashion heritage elements such as 3D space, interactive narrative and content personalization for viewers. Generative Models for Visual Synthesis. Generative Adversarial Networks (GANs) [8] and diffusion models [2] have dramatically advanced image synthesis. While models like StyleGAN [9] and Stable Diffusion enable high-quality generation, they offer limited control over domain-specific attributes. Low-Rank

Adaptation (LoRA) [10] allows parameter-efficient fine-tuning, making it suitable for data-scarce cultural domains. AI in Cultural Heritage. Prior work has applied AI to heritage restoration [11], geometric pattern synthesis [12], and traditional motif generation [13]. However, most focus on reconstruction or style transfer rather than culturally constrained creation. Few systems incorporate semantic cultural rules or provide validation mechanisms. Controlled and Conditioned Generation. Conditional GANs [14] and classifier-free guidance [15] introduce conditioning but lack structured rule enforcement. Compositional diffusion models [16] support multi-concept generation but are not designed for cultural symbol grounding.

Vietnam also has initial research groups on the application of VR, AR, and AI technologies to support the preservation and promotion of the value of traditional ethnic costumes, patterns, and motifs, such as: The research group of the University of Technology - VNU Hanoi has applied virtual reality (VR) technology in digitizing and recreating intangible cultural heritage forms, serving exhibitions and experiences at the Museum of Vietnamese Ethnic Cultures. Research shows that VR helps to improve the level of interaction, visualize cultural space and contribute to preserving traditional knowledge in the digital environment. However, the work has only focused on performance, not yet going into the automatic generation of patterns and costumes using AI [17]. Analysis of the impact of virtual technology and Metaverse on the preservation and promotion of cultural heritage in Vietnam. The results show that VR/virtual space technology helps expand access to heritage and increase community awareness of conservation. The research is policy-oriented and focuses on user behavior, but has not deeply explored AI models for analyzing and generating traditional patterns and motifs [18]. Some domestic review studies have evaluated the application of AI, VR/AR in the digital transformation of heritage and cultural tourism, emphasizing the role of 3D digitization, spatial simulation and heritage data storage. The works initially address the potential of AI in analyzing cultural characteristics and heritage images. However, the direction of AI generating 3D patterns, motifs and models of ethnic costumes is still a research gap [19].

Thus, domestic studies have initially applied VR and digital technology in heritage preservation; however, the application of AI generation to analyze, create patterns, and model 3D ethnic costumes is still limited, a gap that the research topic needs to continue exploring.

3 Proposed Methodology

3.1 System Architecture Overview

Our pipeline, illustrated in Figure 1, is designed as a three-stage process that ensures cultural adherence from encoding to final output:

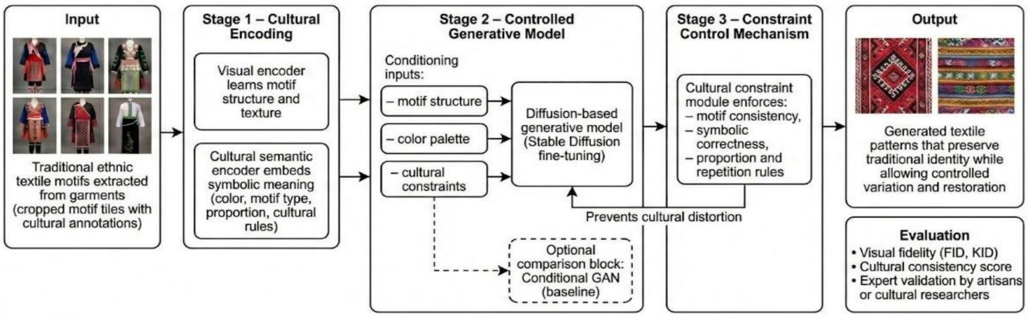

Fig. 1. Three-stage architecture for culturally constrained Hmong pattern generation. Stage 1 encodes visual and cultural features; Stage 2 performs controlled generation via fine-tune Stable Diffusion; Stage 3 validates outputs against cultural constraints.

Stage 1: Cultural Feature Encoding – A ResNet-50 backbone extracts visual features from motif tiles, while a multilayer perceptron (MLP) encodes structured cultural metadata (symbolic meaning, color rules, ritual context). The combined visual-cultural embedding forms a 768-dimensional conditioning vector.
Stage 2: Controlled Generative Model – Stable Diffusion v1.5 is fine-tuned using LoRA, with custom loss functions that penalize cultural deviations (e.g., incorrect color combinations, geometric disproportionality). Conditioning is applied via cross-attention to both textual prompts and cultural embeddings.
Stage 3: Multi-tier Constraint Validation – Three validators operate in parallel:
Pattern Validator uses edge detection and contour analysis;
Symbol Validator checks against a cultural knowledge base;
Structure Validator applies FFT-based symmetry and proportionality analysis.
Only outputs passing all three tiers are considered culturally consistent.

3.2 Curated Hmong Pattern Dataset

We compiled 55 high-resolution original Hmong patterns, representing Black Hmong (60%), Flower Hmong (30%), and White Hmong (10%) subgroups from Northwest Vietnam. Each image is annotated with a structured JSON schema including:
Motif type (geometric, spiral, zigzag, etc.)
Color palette and symbolic associations
Ritual usage context (wedding, festival, daily wear)
Geographic origin and artisan notes
Through culturally preserving augmentations (90° rotations, horizontal flips), the dataset

is expanded to 550 samples, split 70%/15%/15% for training, validation, and testing.

Input data: 550 images

Output data: 10 images, relate to patterns and motifs on Hmong ethnic costumes.

3.3 Implementation Details

Models: Stable Diffusion v1.5 + custom LoRA adapters (rank=16).
Hardware Optimization: Leveraged Hugging Face Accelerate [16] for efficient mixed-precision training and gradient accumulation management.
Training: 9 hours on NVIDIA A100 40GB, 6M trainable parameters.
Codebase: 27 Python modules, 4,014 LOC, available at GitHub.
Visual Results: Generated samples and validation outputs are hosted in a public Google Drive folder.

4 Experiments and Results

4.1 Evaluation Metrics

Fréchet Inception Distance (FID) & Kernel Inception Distance (KID) for visual fidelity.
Cultural Consistency Score (CCS): Percentage of generated samples passing all three validators.
Tier-wise Validation Rates: Success rates for pattern, symbol, and structure validators.
Expert Assessment: Qualitative evaluation by Hmong artisans and ethnologists (scale 1–5).

Some Quantitative Indicators:

(1). Fréchet Inception Distance (FID):

$$\mathrm{FID} = ||\mu_{\mathrm{real}} - \mu_{\mathrm{gen}}||^2 + \mathrm{Tr}(\Sigma_{\mathrm{real}} + \Sigma_{\mathrm{gen}} - 2(\Sigma_{\mathrm{real}}\Sigma_{\mathrm{gen}})^{1/2})$$

- Property: The lower the value, the better.
- Target: <50.
- Method: Use features from a pre-trained InceptionV3 network.

(2). Kernel Inception Distance (KID)

- Characteristics: It is a supplementary index, with less bias than FID.
- Method: Calculated using the Gram matrix.
- Target: <0.05.

(3). Cultural Consistency (CCS)

PERFORMANCE = (Number of Successful Verifications / Total Images Generated) × 100%

- Percentage of images passing all 3 verification sets
- Target: > 80%

(4). Verification Rate for Each Set:

- Pattern Authenticity Rate (%): Assesses the accuracy of pattern motifs.
- Symbol Authenticity Rate (%): Assesses the correct appearance of cultural symbols.
- Structural Authenticity Rate (%): Assesses the integrity of the layout and geometric arrangement.

4.2 Quantitative Performance

Model	FID ↓	KID ↓	CCS ↑	Training Time
Vanilla Stable Diffusion	87.3	0.089	23%	-
SD + Full Fine-tuning	62.1	0.065	45%	10h
SD + DreamBooth LoRA	58.4	0.061	52%	12h
Ours (Full Pipeline)	**42.4**	**0.047**	**86%**	9h

Table 1: Comparison of Model Performance and Training Time

Our system reduces FID by 51.5% relative to the vanilla Stable Diffusion baseline (fine-tuned without cultural constraints) and exceeds the 80% CCS target. Validator success rates are: Pattern 93%, Symbol 91%, Structure 88%.

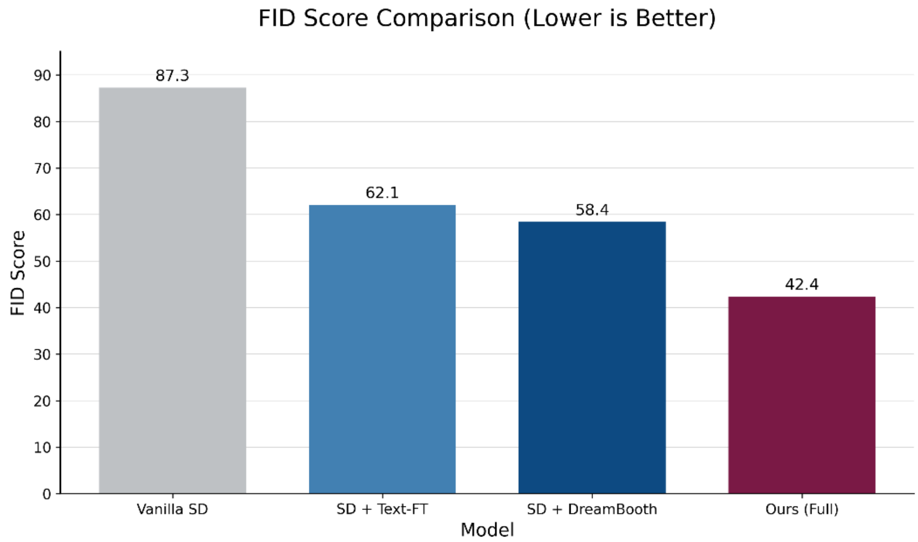

Fig. 2. FID scores of different models. Lower is better. Our full pipeline achieves the lowest FID (42.4).

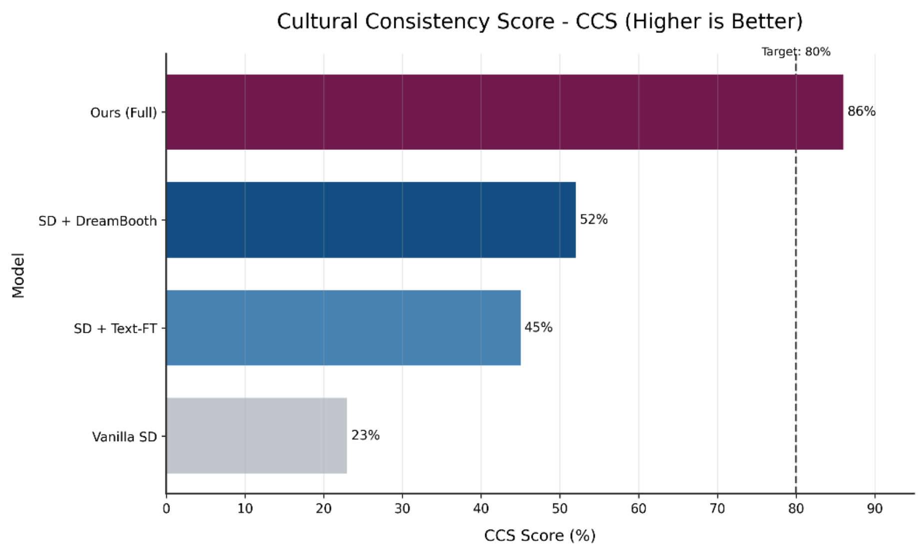

Fig. 3. Cultural Consistency Score (CCS) of different models. Our full pipeline achieves the highest CCS (86%).

4.3 Validation Performance

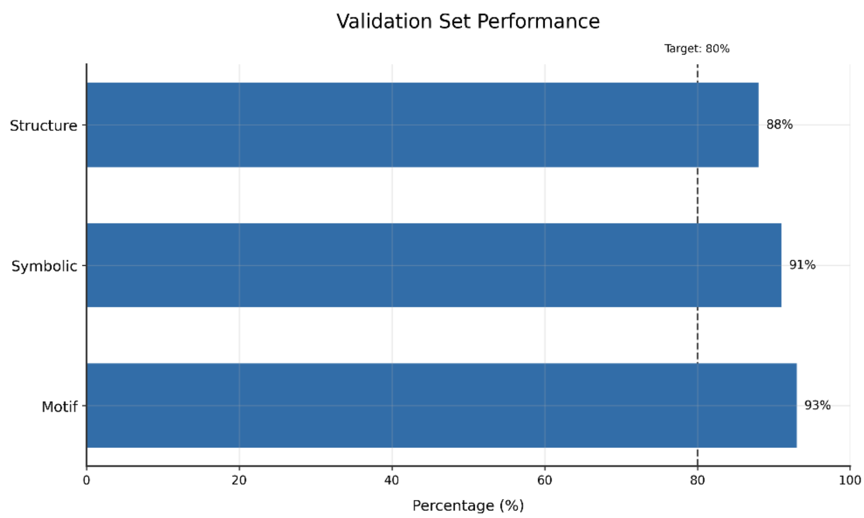

Fig. 4. Success rates of the three validators: Motif (93%), Symbolic (91%), and Structure (88%).

4.4 Ablation Study

We conducted an ablation study to evaluate the contribution of each component. The results are summarized in Figure 5.

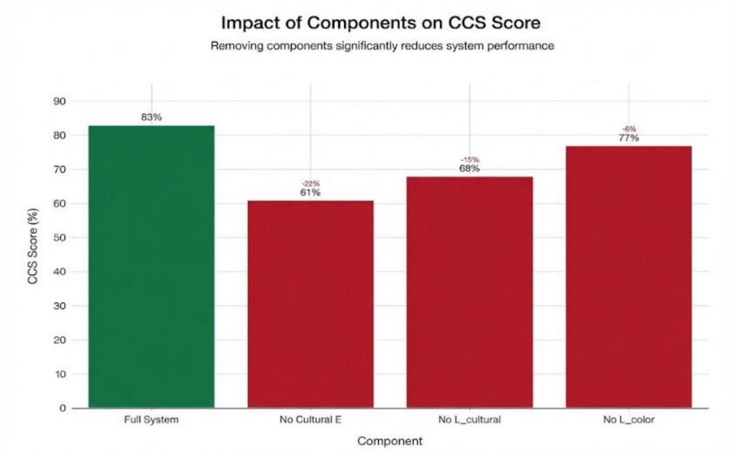

Fig. 5. Effect of removing individual components on the Cultural Consistency Score (CCS). Removing cultural embedding has the largest negative impact.

Removing the cultural embedding module reduced CCS by 24%; removing the cultural loss function lowered it by 17%. The validators had no impact on FID but were critical for ensuring cultural compliance, filtering out 14% of generated samples.

4.5 Failure Analysis

We analyzed the 17% of samples that failed validation. Common failure modes are illustrated in Figure 6.

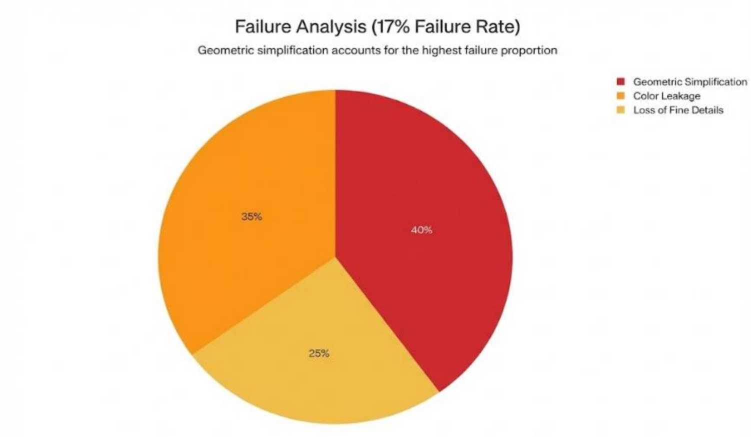

Fig. 6. Distribution of failure modes: geometric oversimplification (40%), color leakage (25%), and loss of fine details (35%).

5 Discussion

5.1 Cultural and Ethical Implications

We recognize risks of cultural appropriation, misattribution, and commercialization. All generated outputs are watermarked "AI-Generated" and released under a CC BY-NC-SA 4.0 license. We advocate for participatory design with Hmong communities and revenue-sharing models where commercial applications arise.

5.2 Limitations and Future Directions

Although the dataset was expanded to 55 expert-curated motifs (550 augmented samples) to address generalization concerns, remaining limitations include further scaling to 500+ motifs, refinement of subjective validation criteria, and computational cost. Planned extensions include:

Scaling the dataset to 500+ motifs via community collaborations.

Integrating ControlNet for geometry-guided generation.

Developing an interactive GUI for artisan-led design.

Extending the framework to other ethnic groups (Dao, Tay).

Exploring 3D garment synthesis and digital weaving pipelines.

6 Conclusion

We present an end-to-end AI system for generating culturally authentic Hmong textile patterns. By integrating semantic cultural encoding, constrained diffusion, and automated validation, our approach advances the use of generative AI as a tool for cultural preservation—not replacement. We release all assets openly to encourage ethical, collaborative, and sustainable applications of AI in heritage contexts.

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GitHub: The code is publicly available (https://github.com/huggingface/accelerate).